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Section: IV - DCSAD	Subject: Elective 4

Tasks:

1. Create a new branch:

- Create a new branch named "**feature_x**" to isolate your experimental changes.

```
Vin@Vin MINGW64 ~/Blanco_local_project (master)
$ git checkout -b feature_x
Switched to a new branch 'feature_x'
```

2. Stage changes:

- Make modifications to your project files.
- Stage the changes using the git add command.

```
Vin@Vin MINGW64 ~/Blanco_local_project (feature_x)
$ echo "Adding experimental features from feature_x" >> main.txt
git add main.txt
warning: in the working copy of 'main.txt', LF will be replaced by CRLF the next
time Git touches it
```

3. Commit changes:

- Commit the staged changes to your local repository using the git commit command. Provide a clear and concise commit message.

```
Vin@Vin MINGW64 ~/Blanco_local_project (feature_x)
$ git commit -m "Update main.txt with experimental feature"
[feature_x bedf492] Update main.txt with experimental feature
1 file changed, 1 insertion(+)
```

4. Switch back to the main branch:

- Use the git checkout command to switch back to the main branch.

```
Vin@Vin MINGW64 ~/Blanco_local_project (feature_x)
$ git checkout master
echo "Adding new code directly to the master branch" >> main.txt
git add main.txt
git commit -m "Update main.txt directly on master"
Switched to branch 'master'
warning: in the working copy of 'main.txt', LF will be replaced by CRLF the next
time Git touches it
[master 4ab578f] Update main.txt directly on master
1 file changed, 1 insertion(+)
```

5. Merge changes from the feature branch:

- Use the git merge command to merge the changes from the "feature_x" branch into the main branch. Resolve any merge conflicts that may arise.

```
Vin@Vin MINGW64 ~/Blanco_local_project (master)
$ git merge feature_x
Auto-merging main.txt
CONFLICT (content): Merge conflict in main.txt
Automatic merge failed; fix conflicts and then commit the result.

Vin@Vin MINGW64 ~/Blanco_local_project (master|MERGING)
$ echo "Resolved line combining both master and feature_x changes" > main.txt
git add main.txt
git commit -m "Merge feature_x into master and resolve conflict"
warning: in the working copy of 'main.txt', LF will be replaced by CRLF the next
time Git touches it
[master c0ca023] Merge feature_x into master and resolve conflict
```

6. Viewing the Commit History:

- Use Git commands to view the commit history and ensure that all commits are in place.

```
Vin@Vin MINGW64 ~/Blanco_local_project (master)
$ git log --oneline --graph --all
*    c0ca023 (HEAD -> master) Merge feature_x into master and resolve conflict
| \
|  * bedf492 (feature_x) Update main.txt with experimental feature
* | 4ab578f Update main.txt directly on master
|/
* 9a4d110 Initial commit
```

7. Tagging Versions (Optional):

- After completing the merge, use the git tag command to create a tag named v1.0 to mark the first version of your project.

```
Vin@Vin MINGW64 ~/Blanco_local_project (master)
$ git tag v1.0
```